

Yanfei Zhu

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EDUCATION

University of Wisconsin-Madison

Madison, WI

VISP, M.A. in Mathematics, GPA: 3.844/4.000

Sep. 2024 – May 2026

Visiting International Student Program (VISP)

- **Courses:** Introductory Topology (1&2, Algebraic Topology), Homological Algebra, Complex Analysis, Abstract Algebra, Ordinary Differential Equations, Modern Algebra, Analysis II

Southern University of Science and Technology (SUSTech)

Shenzhen, CN

B.S. in Mathematics and Applied Mathematics

Sep. 2021 – Jun. 2025

- **Weighted Average Score:** 88.09/100
- **Courses:** Algebraic Topology, Topology, Representation of Groups, Abstract Algebra, Advanced Linear Algebra, Real Analysis, Complex Analysis, Mathematical Analysis, Partial Differential Equations, Ordinary Differential Equations, Discrete Mathematics, Elementary Number Theory, Probability and Statistics, Calculus

Research Interests: I am interested in using algebraic and topological methods to study geometric problems, especially those involving manifolds. My current research focuses on equivariant homotopy theory and homological stability. I am also interested in optimization on manifolds.

RESEARCH AND READING EXPERIENCE

Directed Reading in Manifold Optimization

Jun. 2026 – Present

Advisor: Prof. Qinglin Tang, Sichuan University

Topics: Manifold optimization

Directed Reading in Intersection Homology

Aug. 2025 – Dec. 2025

Advisor: Prof. Laurentiu Maxim, Mathematics Department, UW–Madison

Topics: Intersection homology, sheaves on topological spaces, Verdier duality

Directed Reading in Riemannian Geometry and Morse Theory

Mar. 2025 – Oct. 2025

Advisor: Prof. Ruobing Zhang, Mathematics Department, UC San Diego

Topics: Riemannian geometry (Petersen, do Carmo), Morse theory (Milnor)

Directed Reading in Homological Stability

Apr. 2025 – Jul. 2025

Advisor: Prof. Yifei Zhu, Mathematics Department, SUSTech

Topics: Homological stability, configuration spaces, FI-modules, braid groups

Undergraduate Thesis

Dec. 2024 – Jul. 2025

Symmetry of Solutions in Nonlinear Aggregation-Diffusion Equations and Its Applications

Advisor: Prof. Linlin Su, SUSTech

Studied symmetry properties of stationary solutions to nonlinear aggregation-diffusion equations, with a focus on radial symmetry and applications to numerical methods.

Reading Seminar in Homological Algebra

Sep. 2023 – Dec. 2023

Advisor: Prof. Ziqing Xiang, Mathematics Department, SUSTech

Topics: Category theory, homological algebra

Preprints

The $RO(C_q)$ -Graded Cohomology of Complete Flags in the Regular Representation

Preprint, 2026.

Computes the genuine equivariant cohomology ring, Schubert products, and fixed-point restriction for complete flags in the complex regular representation of C_q , for odd primes q .

Two Lifted Actions on Oriented Configuration Spaces: Stabilization and Bredon Homology

Preprint, 2026.

Finite generation of integral Bredon homology for oriented configuration spaces, using both lifts of a reflection action.

TEACHING EXPERIENCE

Grader, Department of Mathematics, UW–Madison

Feb. 2026 – May 2026

Graded undergraduate mathematics assignments and provided written feedback.

SUMMER SCHOOLS AND WORKSHOPS

IWoAT Summer School, Chinese Academy of Sciences

Aug. 2026

Beijing, CN. Topics: Stable homotopy theory and brave new algebra.

IWoAT Summer School, Westlake University

Jul. 2025

Hangzhou, CN. Topics: Adams spectral sequence computations; the last Kervaire invariant problem.

Summer School, University of Science and Technology of China

Jul. 2025

Hefei, CN. Topics: Geometry and analysis.

SKILLS AND LANGUAGES

Technical Skills: LaTeX, Python, MATLAB, Java

Languages: Mandarin (native), English (fluent), Japanese (basic)